

Two-Phase Locking Protocol can ensure the conflict serializability of concurrent transactions.

选择一项:

- ☒ 对 ✓
- ☐ 错

Given a relation scheme $SC(Sno, Cno, Grade)$, which of the following functional dependencies are non-trivial ()

选择一项:

- ☐ A. $(Sno, Cno) \rightarrow Cno$
- ☐ B. $(Sno, Cno) \rightarrow Sno$
- ☐ C. $Sno \rightarrow Sno$
- ☒ D. $(Sno, Cno) \rightarrow Grade$ ✓

Durability of transactions requires ()

选择一项:

- ☐ A. Either all or none of the operations in a transaction are executed.
- ☐ B. The execution of one transaction should not interfere with another.
- ☐ C. Only the results from the successfully executed transactions are included in the database.
- ☒ D. All the updates to the data should be made permanent in the database. ✓

To normalize a 3NF relation scheme to BCNF, we need to ()

选择一项:

- ☐ A. eliminate the trivial functional dependencies
- ☐ B. eliminate the transitive functional dependencies and partial functional dependencies of prime attributes on any candidate key
- ☐ C. eliminate the partial functional dependencies of non-prime attributes on a candidate key
- ☒ D. eliminate the transitive functional dependencies of non-prime attributes on a candidate key ✗

If $R \in 3NF$ and R has one candidate key, then $R \in BCNF$. ()

选择一项:

- ☒ 对 ✓
- ☐ 错

In a database system with checkpoint technology, the correct order of following operations (① Write database, ② Write log, ③ Write checkpoint record, ④ Write restart file) should be: ()

选择一项:

- ☐ A. ④-③-②-①
- ☒ B. ②-③-①-④ ✓
- ☐ C. ①-②-③-④
- ☐ D. ①-④-③-②

The concurrent execution of transactions are correct, if and only if the concurrent execution results are the same as the results from sequential execution of these transactions in certain order.

选择一项:

- ☒ 对 ✓
- ☐ 错

Suppose there are the following transactions: T1: Committed before checkpoint; T2: Execution starts before checkpoint, and commits before failure but after the checkpoint. T3: The execution starts before the checkpoint and is not completed at the time of failure. T4: Starts execution after checkpoint and commits before the failure point. T5: The execution starts after checkpoint and is not completed at the time of the failure. When using checkpoints for recovery, UNDO is required for: ()

选择一项:

- ☐ A. T1, T2
- ☐ B. T2, T4
- ☒ C. T3, T5 ✓
- ☐ D. T2, T3

If a transaction T has obtained a x-lock on data item Q. Which operations can T do on Q: ()

选择一项:

- ☒ A. read and write ✓
- ☐ B. write only
- ☐ C. read only
- ☐ D. neither read nor write

In a relation scheme, if all the X in the set of functional dependencies ($X \rightarrow Y$) are candidate keys, then the maximum norm form that R can belong to is ()

选择一项:

- ☐ A. 2NF
- ☐ B. 1NF
- ☒ C. BCNF ✓
- ☐ D. 3NF

Which one of the following statements about the recovery of transactions from system failure is correct: ()

选择一项:

- ☐ A. Logs are used to record the frequency of transaction executions.
- ☐ B. In case differential backup is used, there is no need of log files for the recovery.
- ☒ C. Setting up checkpoints for log files can improve the efficiency of recovery. ✓
- ☐ D. Only redo is needed.

Suppose there are the following transactions: T1: Committed before checkpoint; T2: Execution starts before checkpoint, and commits before failure but after the checkpoint. T3: The execution starts before the checkpoint and is not completed at the time of failure. T4: Starts execution after checkpoint and commits before the failure point. T5: The execution starts after checkpoint and is not completed at the time of the failure. When using checkpoints for recovery, REDO is required for: ()

选择一项:

- ☐ A. T5
- ☐ B. T1
- ☒ C. T2, T4 ✓
- ☐ D. T3, T5

Inappropriate concurrent control of transactions may lead to problems. One problem is the loss of updates that have been done. The reason causing this problem is the concurrent execution violates the () of transactions.

选择一项：

- ☐ A. Durability
- ☐ B. Atomicity
- ☐ C. Consistency
- ☒ D. Isolation ✓

Two-Phase Locking Protocol can prevent deadlocking.

选择一项：

- ☐ 对
- ☒ 错 ✓

All or none of the operations in a transaction are executed. What property is it? ()

选择一项：

- ☐ A. Durability
- ☐ B. Consistency
- ☐ C. Isolation
- ☒ D. Atomicity ✓

In a relation scheme R, the meaning of functional dependency $X \rightarrow Y$ is ()

选择一项：

- ☐ A. for any legal relation in R, the value of X should equal to the value of Y
- ☒ B. for any legal relation in R, if the values of X in two tuples equal to each other, then the values of Y should also equal to each other. ✓
- ☐ C. for some legal relation in R, if the values of X in two tuples equal to each other, then the values of Y should also equal to each other.
- ☐ D. for some legal relation in R, the value of X should equal to the value of Y

When a transaction T obtains a X-Lock on data item Q, which operations can other transactions do on Q: ()

选择一项：

- ☒ A. neither read nor write ✓
- ☐ B. read but not write
- ☐ C. read and write
- ☐ D. write but not read

Any relation scheme containing two attributes must belong to BCNF ()

选择一项：

- ☐ 对
- ☒ 错 ✗

If a relation scheme R does not have a non-prime attribute, then R must ()

选择一项：

- ☒ A. belong to BCNF ✗
- ☐ B. not belong 2NF
- ☐ C. belong to 2NF but not necessarily 3NF
- ☐ D. belong to 3NF but not necessarily BCNF